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10^{-21} -Level optical frequency dissemination over 2067 km of noise-loaded field-deployed fiber network

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Abstract

Achieving ultra-stable optical frequency dissemination over long-haul fiber networks is essential for numerous applications. Although optical frequency transfer systems based on optical phase-locked loops (OPLLs) have achieved unprecedented stability levels, their performance is limited by continuous compensation bias caused by noise asymmetry from bidirectional frequency shifts and loss of lock in long-distance, high-noise links. Here, we propose a bias-free noise compensation method based on digital radio-frequency phase recording using a time-to-digital converter, which eliminates residual errors and offers a theoretically unlimited dynamic range for enhanced reliability. By incorporating multifunctional relay stations and hertz-level optical bandpass filtering to improve OPLL robustness, our scalable architecture achieves a frequency instability of 2.9×10^{-21} at 1 day over a 2067 km field fiber link under extreme noise conditions ($5000 \text{ rad}^2 \text{ Hz}^{-1} \text{ km}^{-1}$ at 1 Hz). Bias correction improves the instability by threefold and breaks through the theoretical limit of uncalibrated systems. The setup maintains continuous phase lock for over four days, and noise purification enables virtually unlimited link extension. This advance establishes a robust, field-deployable optical frequency network compatible with standard telecommunication infrastructure.

Introduction

The global dissemination of ultra-stable optical frequencies over fiber networks represents a transformative capability at the forefront of precision metrology^{1–3}, forming the critical infrastructure for a new era of scientific exploration. This technological advancement enables a new generation of groundbreaking experiments, including intercontinental optical clock comparisons^{4–7}, very-long-baseline interferometry^{8,9}, next-generation geodesy^{10,11}, and rigorous tests of fundamental physics^{12–14}.

Central to this advancement is the optical phase-locked loop (OPLL), which actively suppresses the pervasive

phase noise induced by transmission fibers. Through continuous refinement, OPLL-based optical frequency dissemination (OFD) systems have achieved remarkable fractional frequency instabilities at the 10^{-18} to 10^{-20} level over intracontinental distances^{4,15,16}, marking a significant milestone in frequency metrology. However, performance remains far from reaching the fundamental limit due to residual noise from imperfect cancellation and the frequent loss of lock.

First, the OPLL operates under the assumption that round-trip phase noise is symmetric, enabling noise cancellation by compensating for half of the total round-trip phase noise—this yields zero deviation when the forward (ν_f) and the backward (ν_b) optical signals share the same frequency¹⁷. In practice, however, fiber links are subject to significant parasitic reflections. All long-distance optical frequency transfer employs frequency shifting at the remote end to tag the signal, exploiting the resulting frequency difference ($\Delta\nu$) to suppress interference from single-pass parasitic reflections^{18,19}. Nevertheless, $\Delta\nu$

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disrupts the symmetry required for ideal noise cancellation in the OPLL, leading to a residual noise on the order of 10^{-7} , i.e., $(1/\nu_f - 1/\nu_b)/2 \approx \Delta\nu/2\nu_0$. Importantly, as long as the phase compensation value is recorded, this deviation can be calculated and corrected.

On the other hand, OPLLs frequently lose lock under high-noise conditions^{5,15,16,20}, and although anomalies can be removed afterward, this still degrades link stability. To date, only one low-noise OFD link exceeding 1000 km has been reported to operate continuously for several days without losing lock²¹. The OPLL is a typical negative feedback system, and the loss of lock typically occurs when excessive system noise causes the feedback error signal to exceed the monotonic dynamic range, rendering the feedback ineffective. OPLL robustness can therefore be enhanced by expanding the monotonic dynamic range and/or reducing the noise level.

Conventional phase discriminators, such as RF mixers, rely on zero-phase cross-correlation between reference and detected heterodyne signals and are constrained to a monotonic dynamic range of π radians (half cycle)¹⁷. Current approaches to extend this range include frequency division of the optical beat note prior to phase discrimination^{4,5,15,22} and the use of digital phase-unwrapping algorithms²³. Nevertheless, to date, the maximum achieved monotonic dynamic range remains below 1000π radians of optical phase²², which is insufficient for many applications. Although digital unwrapping algorithms can theoretically extend the dynamic range indefinitely²³, they often fail under fast phase transient conditions due to error propagation and limited sampling rates. Enhancing the OPLL's monotonic dynamic range alone cannot fully prevent frequent lock loss.

Cascading OFD links is an effective method for extending link length and reducing OPLL noise levels. However, in very long cascaded configurations, accumulated noise across multiple stages can still compromise system robustness. This issue is further exacerbated by the fact that each OFD link typically has a similar span²², causing its OPLL gain peaks to align at approximately the same frequency, thereby amplifying the overall noise accumulation. While noise suppression techniques based on ultra-stable laser tracking have been proposed and experimentally demonstrated¹⁶, the practical effectiveness of parallelization strategies over long-distance links has not yet been experimentally validated.

In this work, we present a scalable OFD system that not only corrects bias in fiber noise compensation but also improves robustness by extending the monotonic dynamic range of the OPLL and suppressing optical phase noise. We achieve high-resolution optical phase recording with sub-radian precision over a theoretically unlimited range by directly measuring the phase of the radio frequency (RF) optical beat note through precise time-of-

flight measurements of digitally conditioned RF signals. Since noise is detected in the digital domain, we enable real-time link noise compensation using bias-corrected phase information. Furthermore, we implement a hertz-level optical bandpass filter to mitigate accumulated phase noise in long and noisy cascaded links. To validate the proposed techniques, we deployed the developed system in a field experiment to establish a 2067-km round-trip cascaded OFD link. The fiber link exhibits a high noise density of approximately $5000 \text{ rad}^2 \text{ Hz}^{-1} \text{ km}^{-1}$ at 1 Hz. Nevertheless, the system achieves a fractional frequency instability of 2.9×10^{-21} at 1 day, incorporating an optical phase noise filter at the 1267-km node. Notably, the phase noise density of the full 2067-km link is lower than that of the 1267-km segment, demonstrating that the link can be extended to arbitrary lengths when equipped with intermediate optical filtering stages.

Results

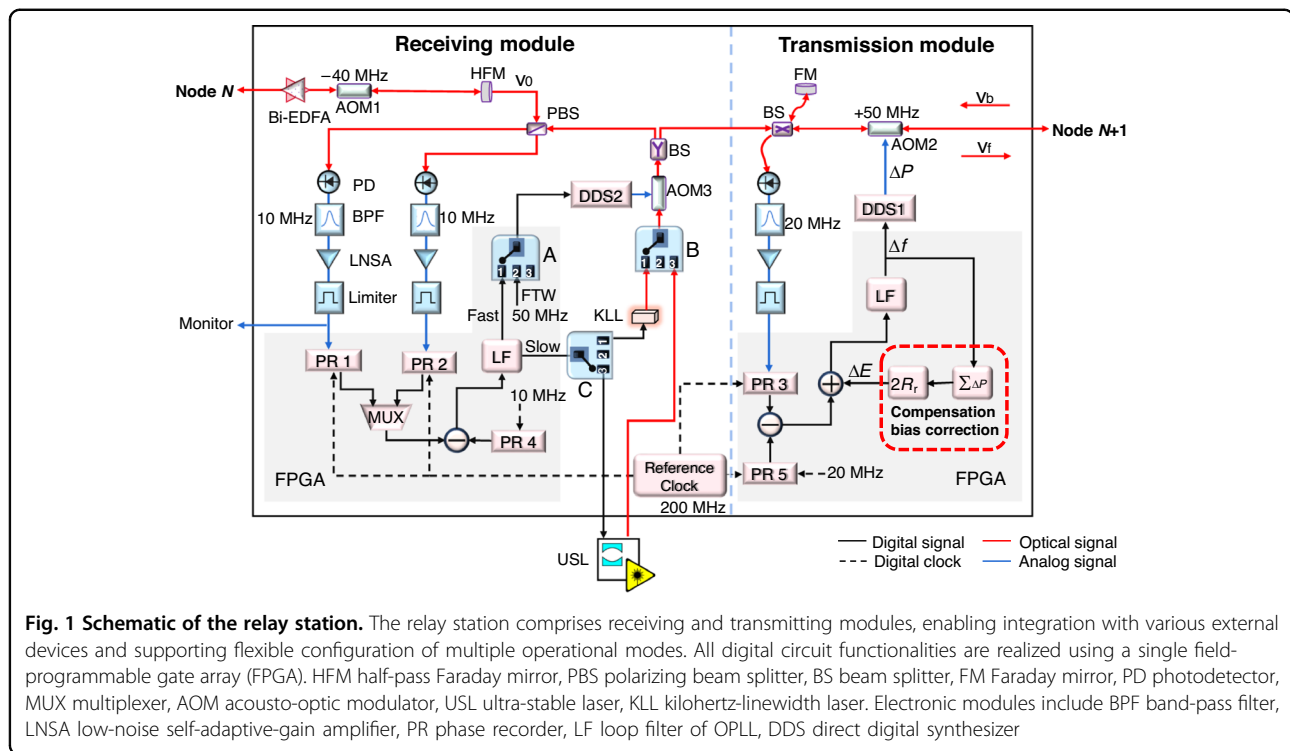
Development of ultra-stable long-haul OFD links

Building upon the previously introduced conceptual framework, we have developed a long-distance OFD system centered on relay station units, capable of real-time, bias-free noise compensation and robust operation in high-noise field-deployed fiber environments. The system comprises three components: relay stations, bidirectional optical amplifiers, and ultra-stable lasers with 1555.12 nm wavelength and sub-Hertz linewidth. Relay stations are equipped with real-time fiber noise suppression functionality, support cascaded deployment, and feature dedicated ports for external reference signal input. The ultra-stable laser provides a highly coherent optical signal at the designated frequency. By employing a loosely phase-locked control scheme, the relay stations thus actively suppress phase noise from the incoming carrier laser, effectively mitigating accumulated phase disturbances that would otherwise lead to OPLL unlocking over extended distances. Furthermore, both integrated and external bidirectional optical amplifiers within the relay stations enhance the signal-to-noise ratio (SNR) of the detected optical carrier, compensating for attenuation-induced degradation due to fiber transmission loss. Relay stations and bidirectional optical amplifiers support remote parameter configuration and data acquisition.

To validate the performance of the OFD system, we conducted field tests over a 2067-km round-trip optical fiber link connecting Shanghai to Taihu County, Anqing City, Anhui Province, employing our self-developed equipment. Key performance metrics, including link noise and system stability, were accurately calibrated.

Relay station for scalable OFD

The relay station (Fig. 1) serves as the central component of our scalable OFD architecture, capable of flexibly

**Table 1 Settings of relay station modes**

Switch A	Switch B	Switch C	Relay station mode
2	3	2	Initial node
1	1	1	Cascaded node
1	3	3	Cascaded noise purification node

switching among multiple operational modes through switch selection and parameter configuration. The mapping between switch states and corresponding operating modes is shown in Table 1.

The processing flow and control logic for the laser signal in the relay station differ across various operational modes. In the initial node mode, an external reference laser is injected into the relay station via a dedicated interface and directly routed to the transmission module through an acousto-optic modulator (AOM3). In this configuration, the direct digital synthesizer (DDS2) driving AOM3 is set to operate in fixed-frequency output mode. In the cascaded mode, a “dual-dimensional control” strategy is implemented to enable cascaded locking and regeneration of the laser signal. Here, the local reference laser, with a linewidth on the kilohertz scale, is phase-locked to the incoming signal from the upstream station using an OPLL. During the locking process, coarse frequency adjustments over a wide range are performed via the laser’s built-in piezoelectric transducer (PZT),

while fine, broadband tuning at the kilohertz level is achieved through DDS2. Once phase lock is established and the regenerated signal reaches stable operation, it is delivered to the transmission module. In the cascaded purification mode, the primary objective shifts to signal purification. To significantly improve frequency stability, the local laser source is switched to an ultra-stable laser with a sub-hertz linewidth. The OPLL-based phase-locking mechanism remains in use, but the control parameters are tailored to the characteristics of the ultra-stable laser: DDS2 performs loose phase locking within a hertz-level bandwidth, while large-range frequency adjustments are executed through the laser’s native digital frequency modulation interface to maintain long-term lock integrity. The purified signal is then fed into the transmission module, enabling the cascaded distribution of low-phase-noise optical signals.

The transmission module is specifically designed for fiber noise compensation, with a Michelson interferometer serving as its core component. This interferometer features asymmetric arm lengths and polarization insensitivity, enabling high-precision measurement of the additional phase noise accumulated by the optical signal during round-trip propagation through the fiber. Structurally, the short arm terminates at a Faraday mirror (FM), which provides a stable phase reference for noise detection, while the long arm acts as a signal transmission channel, delivering the optical signal to the next relay station. Upon arrival at the subsequent

station, the signal undergoes frequency shifting via AOM1 before being reflected by a half-pass Faraday mirror (HFM), thus completing the round-trip path. The fiber-induced noise is embedded within the heterodyne RF signal captured by the photodetector integrated in the interferometer. This signal is processed by a field-programmable gate array (FPGA)-based phase recorder that exhibits a wide and monotonic dynamic range in the OPLL, thereby facilitating the calibration of noise compensation deviations. Phase noise compensation is achieved by adjusting the output phase of DDS1—the driver of acousto-optic modulator AOM2—thereby actively suppressing the measured phase noise. A dedicated FPGA-based logic module is incorporated into the system to perform real-time computation and feedback control, ensuring high-precision phase compensation.

The receiving module is designed to regenerate the optical carrier signal transmitted from the preceding station, accomplished by employing an OPLL to lock the local laser to the incoming signal. The control reference signal, analogous to that in the transmission module, is acquired via RF phase recorders from the optical heterodyne interference signal, with the control scheme implemented as previously described. A critical challenge arises from polarization state fluctuations in the out-of-loop RF signal used for laser locking, which can severely degrade the SNR of the beat note and subsequently lead to OPLL unlock events. To mitigate this issue, a polarization beam splitter (PBS)-based heterodyne interferometer is deployed at each receiving module²⁴. This subsystem enables real-time monitoring of orthogonal polarization components and dynamically selects the channel with relatively high SNR, thereby ensuring robust OPLL operation.

Within the relay station, all frequency signal generation and measurement operations are referenced to a common local RF oscillator. Furthermore, the sum of all optical frequency shift components at the station is constrained to zero (i.e., ν_0 is identical in each node), thereby relaxing the stability and accuracy requirements on the local RF reference²².

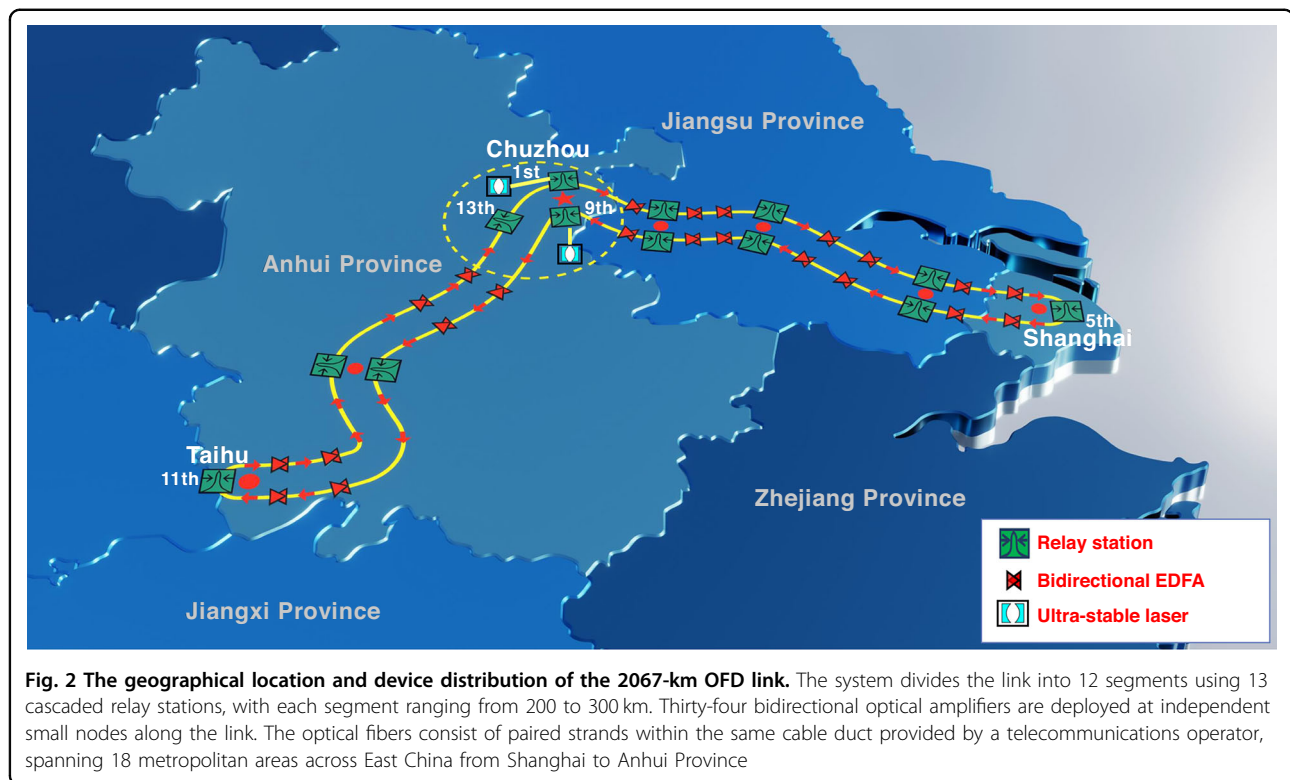
Bias-free noise compensation

To achieve accurate noise compensation, the OPLL incorporates two key features: (1) the integration of digital phase noise measurement (the phase recorder) and phase compensation (the DDS), which enables precise determination of the required compensation value to support bias calibration; and (2) a dedicated digital compensation calibration functional module (Located within the red dashed box in Fig. 1).

RF phase recorders based on time-to-digital converters (TDC)²⁵ are employed to measure the fiber-induced phase noise. The timing reference is derived from the common

frequency reference (200 MHz). As illustrated in Fig. 1, a high-speed photodetector first converts the optical heterodyne beat note into an electrical signal, which is then passed through a band-pass filter (20 MHz) to suppress out-of-band noise components. Subsequently, a low-noise, adaptive-gain amplifier stabilizes the signal amplitude to ensure sufficient drive capability for downstream digital logic circuits, facilitating the generation of a clean square wave. This step effectively minimizes the influence of digital quantization noise and maintains high fidelity in the digital signal domain. Then, a coarse time interval is generated by aligning the rising edge of the input signal with the nearest edge of the reference clock. Subsequently, this signal pair is fed into the TDC for high-precision time interpolation using an FPGA-based time chain architecture, achieving sub-nanosecond resolution^{25,26}, which corresponds to sub-femtosecond precision at optical frequencies. When the resolution of the signal phase measurement is much smaller than the period of the signal, the accuracy of the period during phase measurement can be guaranteed. In the system, the beat frequency signal is 20 MHz, that is, the period time is 50 ns, corresponding to approximately 5 fs of optical phase change. When the resolution of the TDC measurement is 1 ns, the single optical phase measurement resolution is 0.1 fs. The final measured time value, T_m , is obtained by combining the coarse time with the fine time interpolation result. Concurrently, the same time measurement process is applied within the FPGA to the internal reference frequency signal to yield the reference time T_r . Finally, the optical phase error is calculated as $\Delta\varphi = 2\pi \times (T_m - T_r)/T_0$, where T_0 denotes the period of the heterodyne signal, approximately 50 ns. Note that the high-rate digital phase recording eliminates cycle ambiguity entirely, thereby enabling robust operation of the OPLL even under extremely noisy conditions.

As previously mentioned, the noise compensation deviation is determined by the round-trip optical frequency difference (80.0 MHz) and the optical carrier frequency (193.4 THz), yielding a deviation coefficient $R_r = 2.07 \times 10^{-7}$. Since all signals involved are digital, the corresponding compensation term can be efficiently computed and applied in real time. The input to the bias compensation module is the phase correction quantity ΔP generated by the OPLL loop filter to drive the DDS, which is calculated as the product of the DDS frequency increment Δf and the control cycle duration T (100 μ s). Subsequently, the accumulated phase compensation $\sum \Delta P$ used for link noise suppression is multiplied by twice the residual coefficient R_r to produce the error offset ΔE , given by $\Delta E = 2R_r \sum \Delta P$. This offset is then fed back into the subsequent control stage (Fig. 1), thereby completing the correction of the bias noise compensation arising from the bidirectional laser frequency difference.



The performance improvement of bias-free compensation technology can be obtained by comparing and analyzing the link instability results with and without the compensation function enabled. The specific results are presented in the “Relative instability of the OFD link” section.

Construct a 2067-km field OFD link

As shown in Fig. 2 Chuzhou was selected as both the starting and terminating point of the link due to its favorable deployment conditions. To mitigate the impact of high fiber noise, a cascaded architecture is implemented to reduce the phase noise burden on the OPLLs. The link is segmented into 12 segments using 13 relay stations. The frequency transfer signal is generated by a free-running ultra-stable laser and first propagates east-bound to Shanghai before returning to Chuzhou station, covering a round-trip distance of approximately 1267 km. At the 9th node (Chuzhou), a second free-running ultra-stable laser is integrated to form a cascaded noise purification stage. Subsequently, the signal proceeds west-bound to Taihu station and returns. Since both ends of the link are located at Chuzhou, the output signal is looped back to the origin via a short patch fiber, where it is received and monitored for end-to-end stability assessment.

The total link loss is approximately 600 dB (average attenuation: 0.29 dB km^{-1}), compensated by 34 remotely controlled bidirectional erbium-doped fiber amplifiers

(EDFAs), each providing a gain of 1030 dB. Notably, the 9th node can be flexibly switched between standard cascading mode and cascaded noise purification mode, allowing direct evaluation of the performance of the noise purification unit.

The distance scalability of high-noise OFD links

The environmental noise profile encountered along our field-deployed fiber link presents unprecedented challenges for stable frequency dissemination. Urban segments comprising 9.2 km of aerial fiber infrastructure experience intense traffic-induced vibrations and thermal fluctuations, generating exceptional phase noise levels of $5000 \text{ rad}^2 \text{ Hz}^{-1} \text{ km}^{-1}$ at 1 Hz (Fig. 3a)—values exceeding those reported for dedicated fiber networks by orders of magnitude^{4,5,15,16,20,21,27,28}.

Transient environmental disturbances can cause extreme optical phase changes exceeding $10,000 \pi$ radians within a 1-ms time interval—a situation that typically leads to catastrophic failures in traditional analog OPLLs. Our direct RF phase detection design greatly overcomes these limitations by offering a theoretically unlimited phase tracking range and an amplitude dynamic range $> 60 \text{ dB}$, ensuring long-term lock retention even under these extreme noise conditions.

The phase noise spectra of the OFD link are shown in Fig. 3b. While cascaded noise suppression achieves substantial stabilization, noise continues to accumulate

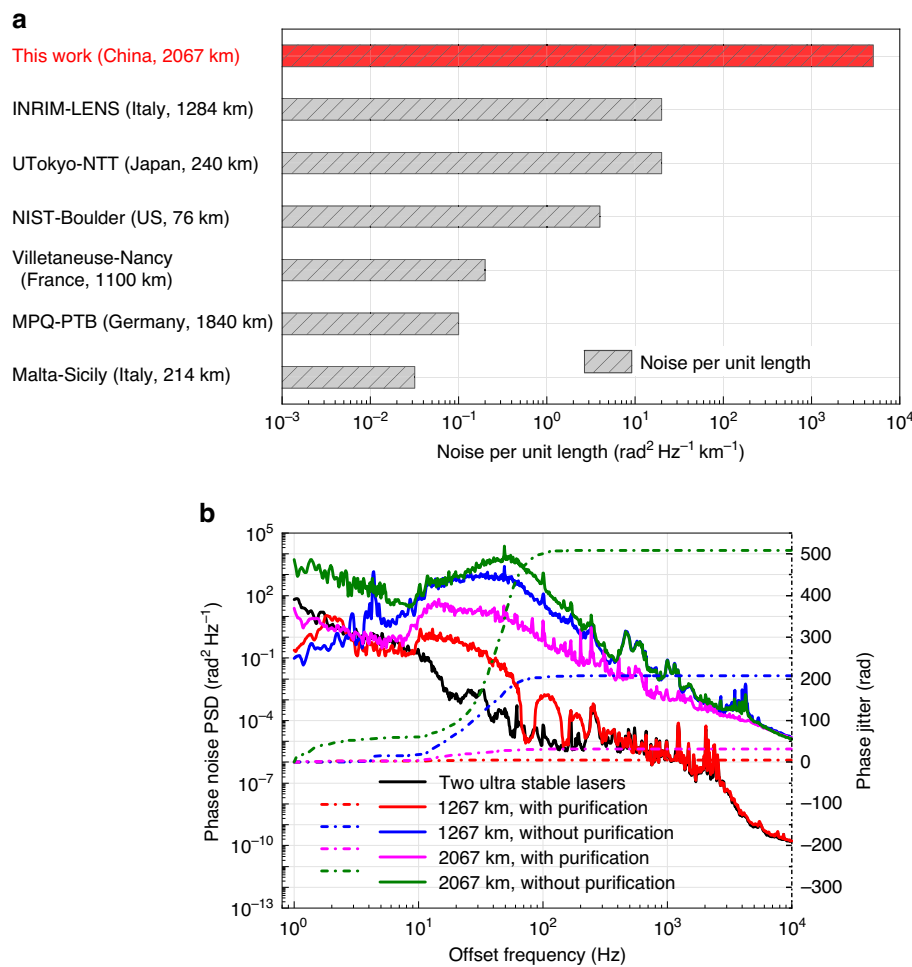
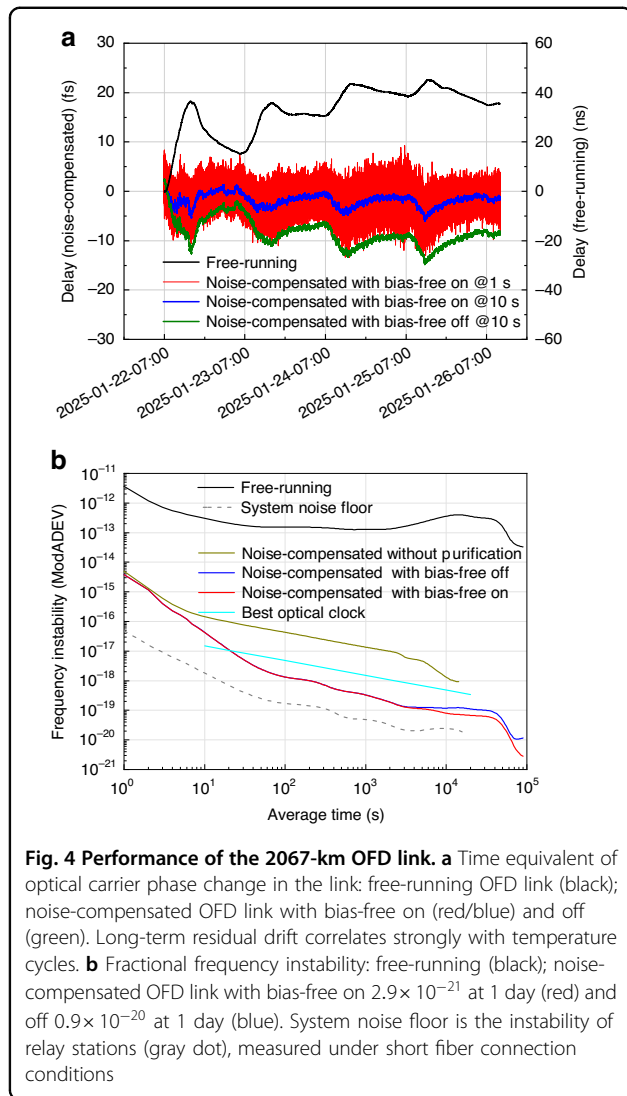


Fig. 3 Phase noise of OFD links. **a** Fiber noise level (at 1 Hz) comparison of OFD links. Malta-Sicily: Italy, submarine fiber links, 214 km²⁷; MPQ-PTB: Germany, 1840 km¹⁵; Villetaneuse-Nancy: France, 1100 km⁵; NIST-Boulder: US, 76 km²⁸; UTokyo-NTT: Japan, 240 km²⁰; INRIM-LENS: Italy, 1284 km¹⁶. **b** Phase noise power spectral density (PSD) and phase jitter. Phase noise PSD of the beat note between two independent ultra-stable lasers at Chuzhou (black). Noise for the 1267 km link with (red) and without (blue) noise purification. Noise for the full 2067 km link with (magenta) and without (green) purification. The left vertical axis corresponds to PSD (solid lines), and the right one to phase jitter (dashed lines)

exponentially over extended distances. The servo-induced resonance between 10 Hz and 5 kHz significantly amplifies phase noise, clearly visible especially in configurations without a noise purification node (blue and green solid curves in Fig. 3b). Upon activation of the noise purification node deployed at the 1267-km mark, laser phase noise above tens of hertz is suppressed to a level comparable to that at the originating node (red solid curve). Notably, after full transmission over 2067 km, the final output exhibits superior noise characteristics relative to the intermediate 1267-km segment (magenta solid curve). Comprehensive phase noise analysis confirms a 20 dB reduction (green and magenta solid curves in Fig. 3b), phase jitter by more than an order of magnitude (green and magenta short dash-dotted curves in Fig. 3b). These results demonstrate that incorporating a single

purification node can extend the OFD link by at least 800 km, with potential for further extension limited only by practical deployment constraints rather than fundamental physical barriers.

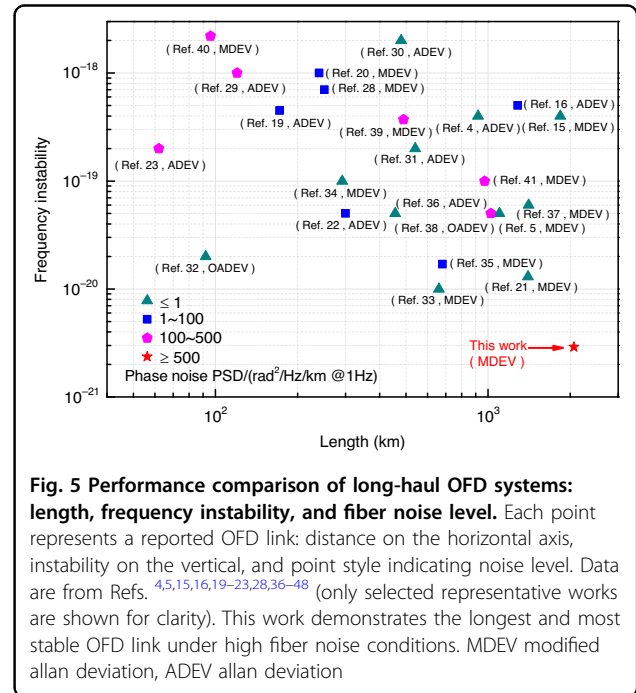
In principle, the length of OFD links can be extended without a fundamental limit by incorporating purification nodes. This is achievable because ultra-stable frequency signals are transmitted here—during the transmission of such signals, only low-frequency noise needs to be preserved to reconstruct the signal at the remote end, while link noise with higher frequencies can be effectively filtered out. To clarify this point, the purification mechanism, transfer function, and simulation verification results are elaborated in the supplementary materials, confirming that intermediate purification can effectively extend the transmission distance of links. It should be noted that



noise within the low-frequency band accumulates to a certain extent, thereby degrading the stability of the links, but this can be further suppressed by adopting technical means such as better ultra-stable lasers and narrower filtering bandwidths.

Relative instability of the OFD link

By comparing the signal phases at both ends of the OFD link, we characterized the system's time drift and frequency instability. The evolution of propagation delay highlights the exceptional noise suppression capability of our architecture (Fig. 4a). The uncompensated link exhibits a pronounced diurnal fluctuation with a peak-to-peak amplitude of approximately 55 ns—recorded via the OPLL compensation signal—that closely follows environmental temperature cycles. Our bias-free compensation scheme reduces the residual time drift to ~6 fs, clearly surpassing the theoretical residual limit of 11 fs (



$2 \times 10^{-7} \times 55$ ns) inherent in conventional noise compensation methods. Data analysis further confirms that disabling bias correction leads to a degradation of time drift to ~15 fs, consistent with theoretical expectations. The remaining ~6 fs residual is primarily attributed to polarization-mode dispersion (PMD) in field-deployed fibers, which represents a fundamental limitation for current OFD systems and will be the focus of our next research phase.

The fractional frequency instability, evaluated using the modified Allan deviation, demonstrates record-breaking performance (Fig. 4b). At an averaging time of one day ($\tau = 1$ day), the instability reaches 2.9×10^{-21} —threefold lower than that achieved with conventional bias compensation techniques.

Thanks to the integration of noise purification, direct RF phase detection, and real-time adaptive correction, our system maintains continuous lock even under long-haul, high-noise conditions (phase noise > 5000 rad² Hz⁻¹ km⁻¹ at 1 Hz). This robustness provides a critical technical foundation for achieving ultra-high stability. A comprehensive comparison with state-of-the-art long-haul OFD systems (Fig. 5) confirms that our approach outperforms all prior demonstrations in both transmission distance and fractional frequency instability, achieving unprecedented precision in optical frequency.

Discussion

We have developed bias-free, cascaded relay stations for OFD links, effectively eliminating the detrimental impact of residual compensation errors historically limiting link

stability. Furthermore, the integrated application of direct optical phase recording via RF extrapolation, noise purification, and real-time adaptive correction ensures exceptional operational reliability of the OFD system. In experimental demonstrations over a 2067-km fiber link, we achieved a fractional frequency instability of 2.9×10^{-21} at an averaging time of one day, while operating under extreme phase noise conditions exceeding $5000 \text{ rad}^2 \text{ Hz}^{-1} \text{ km}^{-1}$ at 1 Hz—surpassing the theoretical performance limits of conventional biased noise compensation schemes. Critically, our approach does not rely on specialized low-noise fibers but instead leverages existing telecommunications infrastructure, a key advantage that significantly expands its feasibility for widespread deployment and diverse scientific and technological applications. The specific links also need to consider complexity, cost, and implementation conditions; in some special cases, such as the frequency transmission of submarine optical cables, more design restrictions may be encountered. We hope that a more systematic solution can be formed in the future.

This advancement addresses the pressing needs of next-generation global precision applications, such as the redefinition of the SI second via optical clock networks^{1,2,4–7}, high-precision relativistic geodesy and seismic sensing^{10,11,29,30}, distributed quantum information networks^{31,32}, enhanced dark-matter detection strategies^{33,34}, and revolutionary experiments in fundamental metrology^{1,2,35}. By bridging the critical gap between laboratory-level stability and practical field deployment, our work establishes the foundation for an intracontinental-scale precision metrology infrastructure that will simultaneously accelerate scientific discovery and drive technological innovation across multiple domains.

Materials and methods

System integration

The experimental implementation of the cascaded fiber noise suppression system is organized into two modular, rack-mounted subsystems: a thermally stabilized optical assembly (1U height) and an electronic control unit (3U height). The optical module integrates all critical passive components—excluding the AOM and laser sources—within a compact, shielded enclosure (size sub- $10 \times 15 \times 5$ cm), providing milli-Kelvin thermal stability to minimize thermo-optic phase drifts in the fiber paths.

A comprehensive digital signal processing chain is implemented on a Xilinx Spartan-6 FPGA, performing real-time fiber noise cancellation, regenerative heterodyne phase stabilization, and advanced phase-domain control. Key functionalities include event-timing-based phase recording, adaptive loop filtering, and correction of phase compensation bias induced by bidirectional propagation asymmetry. Theoretically, the data throughput and

storage requirements for continuous TDC phase recording are relatively large, which may limit the ultra-long-term operation of the system. Instead of directly storing the data from the phase recorders, we store and calculate the difference between the two phase recorders and use the method of cumulative summation of data to replace the storage of each data point, thereby significantly reducing the demand for data throughput and storage. Currently, each relay device's FPGA uses about 2000 slice registers, accounting for less than 20% of the total resources. This monolithic integration on a single chip enhances system reconfigurability, reduces footprint, and improves reliability.

Deployed within standard telecommunications facilities, each noise suppression node operates alongside bidirectional EDFAs and ultra-narrow linewidth lasers. The system incorporates 4G-LTE cellular backhaul connectivity for cloud-based remote management, enabling four key capabilities: (1) software-defined parametric optimization of stabilization loops; (2) real-time telemetry collection for noise suppression performance and laser lock metrics; (3) comprehensive remote configuration, diagnostics, and operational analytics; (4) self-recovery functionality after interruption or loss of lock. This design meets carrier-class requirements for modularity, serviceability, and reliability, ensuring seamless integration into existing optical network infrastructure while maintaining 10^{-21} level frequency instability.

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Author contributions

Q. Zhang, H.-F. Jiang, and J.-W. Pan conceived the experiment. F.-X. Chen, L.-B. Li and J.-P. Chen conceived the study, designed and conducted the experiments, analyzed the results, and wrote the manuscript. K. Zhao designed the remote operation and control software system. J.-Y. Guan and Y. Xu contributed to stabilizing the SNR of the beat frequency signal. L. Hou, F. Zhou, and C.-Z. Peng participated in the quality control, discussions of the manuscript, and maintenance of optical fiber links.

Data availability

All data supporting the main findings of this work are available within the paper, or available from the corresponding author upon reasonable request.

Conflict of interest

The authors declare no competing interests.

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References

- Dimarcq, N. et al. Roadmap towards the redefinition of the second. *Metrologia* **61**, 012001 (2024).
- Lindvall, T. et al. Coordinated international comparisons between optical clocks connected via fiber and satellite links. *Optica* **12**, 843–852 (2025).
- Chen, Z. et al. Time-frequency transfer over optical fiber. *Natl. Sci. Rev.* **12**, nwaf236 (2025).
- Predehl, K. et al. A 920-kilometer optical fiber link for frequency metrology at the 19th decimal place. *Science* **336**, 441–444 (2012).
- Chiodo, N. et al. Cascaded optical fiber link using the internet network for remote clocks comparison. *Opt. Express* **23**, 33927–33937 (2015).
- Koke, S. et al. Validating frequency transfer via interferometric fiber links for optical clock comparisons. *N. J. Phys.* **23**, 093024 (2021).
- Schioppo, M. et al. Comparing ultrastable lasers at 7×10^{-17} fractional frequency instability through a 2220 km optical fibre network. *Nat. Commun.* **13**, 212 (2022).
- Clivati, C. et al. A VLBI experiment using a remote atomic clock via a coherent fibre link. *Sci. Rep.* **7**, 40992 (2017).
- Clivati, C. et al. Common-clock very long baseline interferometry using a coherent optical fiber link. *Optica* **7**, 1031–1037 (2020).
- Marra, G. et al. Ultrastable laser interferometry for earthquake detection with terrestrial and submarine cables. *Science* **361**, 486–490 (2018).
- Marra, G. et al. Optical interferometry-based array of seafloor environmental sensors using a transoceanic submarine cable. *Science* **376**, 874–879 (2022).
- Lisdat, C. et al. A clock network for geodesy and fundamental science. *Nat. Commun.* **7**, 12443 (2016).
- Delva, P. et al. Test of special relativity using a fiber network of optical clocks. *Phys. Rev. Lett.* **118**, 221102 (2017).
- Sanner, C. et al. Optical clock comparison for Lorentz symmetry testing. *Nature* **567**, 204–208 (2019).
- Droste, S. et al. Optical-frequency transfer over a single-span 1840 km fiber link. *Phys. Rev. Lett.* **111**, 110801 (2013).
- Calonico, D. et al. High-accuracy coherent optical frequency transfer over a doubled 642-km fiber link. *Appl. Phys. B* **117**, 979–986 (2014).
- Ma, L. S. et al. Delivering the same optical frequency at two places: accurate cancellation of phase noise introduced by an optical fiber or other time-varying path. *Opt. Lett.* **19**, 1777–1779 (1994).
- Coddington, I. et al. Coherent optical link over hundreds of metres and hundreds of terahertz with subfemtosecond timing jitter. *Nat. Photonics* **1**, 283–287 (2007).
- Jiang, H. et al. Long-distance frequency transfer over an urban fiber link using optical phase stabilization. *J. Opt. Soc. Am. B* **25**, 2029–2035 (2008).
- Akatsuka, T. et al. Optical frequency distribution using laser repeater stations with planar lightwave circuits. *Opt. Express* **28**, 9186–9197 (2020).
- Koke, S. et al. Combining fiber Brillouin amplification with a repeater laser station for fiber-based optical frequency dissemination over 1400 km. *N. J. Phys.* **21**, 123017 (2019).
- Lopez, O. et al. Cascaded multiplexed optical link on a telecommunication network for frequency dissemination. *Opt. Express* **18**, 16849–16857 (2010).
- Hu, L. et al. Performance of digital servos in an optical frequency transfer network. *Rev. Sci. Instrum.* **92**, 053709 (2021).
- Clivati, C. et al. Robust optical frequency dissemination with a dual-polarization coherent receiver. *Opt. Express* **28**, 8494–8511 (2020).
- Kalisz, J. et al. Single-chip interpolating time counter with 200-ps resolution and 43-s range. *IEEE Trans. Instrum. Meas.* **46**, 851–856 (1997).
- Liu, S. X. et al. A low-dead-time FPGA-based time-to-digital converter employing resource-efficient downsampling-multiplexing encoding and dual-histogramming for LiDAR applications. *IEEE Trans. Ind. Electron.* **71**, 13395–13405 (2024).
- Clivati, C. et al. Optical frequency transfer over submarine fiber links. *Optica* **5**, 893–901 (2018).
- Williams, P. A., Swann, W. C. & Newbury, N. R. High-stability transfer of an optical frequency over long fiber-optic links. *J. Opt. Soc. Am. B* **25**, 1284–1293 (2008).
- Grotti, J. et al. Geodesy and metrology with a transportable optical clock. *Nat. Phys.* **14**, 437–441 (2018).
- Oguri, K. et al. Optical-lattice-clock-network technology for gravitational potential sensing. *NTT Tech. Rev.* **21**, 36–42 (2023).
- Yu, Y. et al. Entanglement of two quantum memories via fibres over dozens of kilometres. *Nature* **578**, 240–245 (2020).
- Chen, J. P. et al. Sending-or-not-sending with independent lasers: secure twin-field quantum key distribution over 509 km. *Phys. Rev. Lett.* **124**, 070501 (2020).
- Wcislo, P. et al. New bounds on dark matter coupling from a global network of optical atomic clocks. *Sci. Adv.* **4**, eaau4869 (2018).
- Roberts, B. M. et al. Search for transient variations of the fine structure constant and dark matter using fiber-linked optical atomic clocks. *N. J. Phys.* **22**, 093010 (2020).
- Boulder Atomic Clock Optical Network (BACON) Collaboration. Frequency ratio measurements at 18-digit accuracy using an optical clock network. *Nature* **591**, 564–569 (2021).
- Musha, M. et al. Coherent optical frequency transfer over 50-km physical distance using a 120-km-long installed telecom fiber network. *Opt. Express* **16**, 16459–16466 (2008).
- Terra, O., Grosche, G. & Schnatz, H. Brillouin amplification in phase coherent transfer of optical frequencies over 480 km fiber. *Opt. Express* **18**, 16102–16111 (2010).
- Lopez, O. et al. Ultra-stable long distance optical frequency distribution using the internet fiber network. *Opt. Express* **20**, 23518–23526 (2012).
- Bercy, A. et al. In-line extraction of an ultrastable frequency signal over an optical fiber link. *J. Opt. Soc. Am. B* **31**, 678–685 (2014).
- Raupach, S. M. F., Koczvara, A. & Grosche, G. Optical frequency transfer via a 660 km underground fiber link using a remote Brillouin amplifier. *Opt. Express* **22**, 26537–26547 (2014).
- Kim, J. et al. Optical injection locking-based amplification in phase-coherent transfer of optical frequencies. *Opt. Lett.* **40**, 4198–4201 (2015).
- Guillou-Camargo, F. et al. First industrial-grade coherent fiber link for optical frequency standard dissemination. *Appl. Opt.* **57**, 7203–7210 (2018).
- Husmann, D. et al. SI-traceable frequency dissemination at 1572.06 nm in a stabilized fiber network with ring topology. *Opt. Express* **29**, 24592–24605 (2021).
- Cantin, E. et al. An accurate and robust metrological network for coherent optical frequency dissemination. *N. J. Phys.* **23**, 053027 (2021).
- Clivati, C. et al. Coherent optical-fiber link across Italy and France. *Phys. Rev. Appl.* **18**, 054009 (2022).
- Zhang, X. et al. Coherent optical frequency transfer via a 490 km noisy fiber link. *Chin. Phys. Lett.* **39**, 044201 (2022).
- Liu, L. et al. High-precision optical frequency transfer over a 96 km urban fiber link. *Appl. Opt.* **62**, 683–687 (2023).
- Deng, X. et al. Coherent optical frequency transfer via 972-km fiber link. *Chin. Phys. B* **33**, 020602 (2024).